

Tiered Observation Vectors for Anonymous-DHT Receipts: Contact-Set, Timing, and Congestion Tiers with Conservative MaxL Composition (Series Synthesis)

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Abstract

Several papers in this archive parameterize a “Boss Fight” budget by an *observation model* p_{obs} . In practice, deployments have *multiple* partially observed witness surfaces: e.g., (i) which contacts were queried, (ii) timing/packetization metadata, and (iii) congestion witnesses. If receipts publish only a single scalar, this structure is lost and cross-release comparisons become ambiguous. This note centralizes a minimal, receipt-ready interface: publish an *observation vector* of tiers, each with its own observation probability and per-lookup MaxL (or odds-inflation) bound, together with a conservative rule for combining tiers.

Scope. This is a short interface note. It does not propose new mechanisms; it standardizes how *multiple witness tiers* are published and combined in receipts.

Imports. Trace tier vocabulary and transcript projections are imported from the trace-interfaces note (Synthesis 3).[2] Endpoint metric semantics (MaxL/odds inflation) are imported from the endpoint bridge and odds-inflation paper.[3, 4] Observation thinning (mixture/subsampling) is imported from the observation attenuation note (Synthesis 11).[5] Receipt tuple ownership is imported from the line-item schema (Synthesis 12).[6]

Exports. A compact definition of a *tiered observation vector* and a conservative MaxL composition rule that is stable under the series’ “source-only” publication style.

1 Tiered observation vectors (definition)

A deployment rarely exposes “the transcript” as a monolith. Instead, different adversaries (or the same adversary at different vantages) observe different projections. We model this using tiers.

Definition 1 (Tiered observation vector). Fix a secret C (e.g., queried content label) and a mechanism that induces witness projections $Y^{(1)}, \dots, Y^{(m)}$ as in the trace-interfaces vocabulary.[2] For each tier i , declare:

- a *tier name* and witness definition (what projection is meant),
- a *per-lookup* bound b_i on $\mathcal{L}(C \rightarrow Y^{(i)})$ (or an equivalent odds-inflation label), and
- an *observation probability* $p_i \in [0, 1]$ indicating how often that tier is actually observed (or an auditable upper bound).

Let $Z^{(i)}$ be the thinned observation of tier i with parameter p_i (Synthesis 11).[5] The *tiered observation vector* is $\mathbf{Z} := (Z^{(1)}, \dots, Z^{(m)})$.

Receipt hook. In the line-item schema, this vector lives inside `obs_model`: instead of a single p_{obs} , publish a list of tiers with (p_i, b_i) and the combination rule below.[6]

2 A conservative MaxL composition rule

To keep receipts referee-grade without overfitting to a specific adversary, we adopt a conservative composition rule under a clear, checkable condition.

Definition 2 (Conditional-independence condition). We say tiers are *conditionally independent given C* if, for all c ,

$$\Pr[\mathbf{Z} = \mathbf{z} \mid C = c] = \prod_{i=1}^m \Pr[Z^{(i)} = z_i \mid C = c].$$

This condition is natural when tiers correspond to distinct vantages with independent compromise/visibility events, or when the published tier definitions already condition on shared public side information.

Lemma 1 (MaxL subadditivity under conditional independence). *If $Z^{(1)}, \dots, Z^{(m)}$ are conditionally independent given C , then*

$$\mathcal{L}(C \rightarrow \mathbf{Z}) \leq \sum_{i=1}^m \mathcal{L}(C \rightarrow Z^{(i)}).$$

Proof. By definition, $2^{\mathcal{L}(C \rightarrow \mathbf{Z})} = \sum_{\mathbf{z}} \max_c \Pr[\mathbf{Z} = \mathbf{z} \mid C = c]$. Under conditional independence, this equals $\sum_{\mathbf{z}} \max_c \prod_i \Pr[Z^{(i)} = z_i \mid C = c]$. For each $\mathbf{z}$,

$$\max_c \prod_i a_i(c) \leq \prod_i \max_c a_i(c),$$

so

$$\sum_{\mathbf{z}} \max_c \prod_i \Pr[Z^{(i)} = z_i \mid c] \leq \sum_{\mathbf{z}} \prod_i \max_c \Pr[Z^{(i)} = z_i \mid c] = \prod_i \sum_{z_i} \max_c \Pr[Z^{(i)} = z_i \mid c].$$

Taking $\log_2$ yields the claim. $\square$

Corollary 1 (Receipt-ready tier aggregation). *If tier i has an unthinned bound $\mathcal{L}(C \rightarrow Y^{(i)}) \leq b_i$ and thinning parameter p_i , then Synthesis 11 gives*

$$\mathcal{L}(C \rightarrow Z^{(i)}) \leq \log_2((1 - p_i) + p_i 2^{b_i}).$$

If tiers are conditionally independent given C , then

$$\mathcal{L}(C \rightarrow \mathbf{Z}) \leq \sum_{i=1}^m \log_2((1 - p_i) + p_i 2^{b_i}).$$

Remark 1 (If independence is not defensible). *If a deployment cannot justify the conditional-independence condition, the receipt should publish a joint bound for the combined witness surface (treating $\mathbf{Z}$ as a single tier) rather than using Corollary 1. The schema still applies; only the accountant changes.[6]*

3 Canonical tier names (recommended)

To keep cross-paper citations stable, we recommend three canonical tier classes (a deployment may add others).

- **CSET (contact-set tier):** which overlay contacts were queried / delegated to / responded.
- **TIME (timing tier):** timestamps, packetization/length classes, and timeout/abort surfaces.
- **CONG (congestion tier):** RTT/queueing/loss witnesses governed by the congestion series.

Each tier must map to a trace projection owned by the trace-interfaces note.[2]

4 How this is used elsewhere in the archive (one paragraph)

Boss Fight dial sheets usually publish a single p_{obs} for the CSET tier. Tiered deployments (primary cryptographic path + budgeted DHT fallback) should publish distinct tier vectors per path and combine them using Corollary 1, while treating adaptive fallback selection as an additional witness stream owned by the fallback-tax note.[7, 8]

References

- [1] Anonymous. Receipt Accounting and Horizon Composition for Anonymous-DHT Budgets. Series synthesis note (Synthesis 19), The-One-True-Archive (2026).
- [2] Anonymous. Trace Interfaces for Anonymous DHTs. Synthesis 3, The-One-True-Archive (2026).
- [3] Anonymous. Endpoint Metrics Bridge: Odds Inflation, PML/MaxL, and Identification Sizing. Synthesis 5, The-One-True-Archive (2026).
- [4] Anonymous. Odds Inflation, Privacy Loss, and Endpoint Anonymity Bounds. Anonymity series Paper A, The-One-True-Archive (2026).
- [5] Anonymous. Observation Attenuation for Repeated-Use Privacy Budgets: Mixture/Subsampling Bounds and a Stub-Mix Interface. Synthesis 11, The-One-True-Archive (2026).
- [6] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Synthesis 12, The-One-True-Archive (2026).
- [7] Anonymous. A Tiered Deployment Pattern for Reader Privacy: Primary Cryptographic Path, Budgeted DHT Fallback. Synthesis 10, The-One-True-Archive (2026).
- [8] Anonymous. Adaptive Fallback as a Witness Stream: The Path-Selection Tax and Honest Tiered Receipts. Synthesis 14, The-One-True-Archive (2026).